RUDRESH GAWAS

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PROFESSIONAL SUMMARY

Information Technology engineer (B.E. with Honours in Data Science, 2026; CGPA 7.61/10) with hands-on industry experience at Siemens and a strong project track record across software development, AI/ML, IoT, and data. Delivers end-to-end solutions, from real-time pose estimation and data pipelines to dashboards, automated reporting, and AI-driven classification. Open to full-time opportunities in software engineering, AI/ML, data, and IoT.

EXPERIENCE

Software Developer Intern | Siemens *Goa, India | July 2025 – August 2025*

Real-Time Production Monitoring Dashboard

- Designed and deployed a real-time dashboard integrating live server log data from 3 machine types (AOI SAKI, MEK, SPI SAKI), reducing manual monitoring effort by 70% across the production line.
- Built a data pipeline that processed thousands of machine-generated log files daily, surfacing live production status, machine health, and product insights for floor managers.
- Tech stack: Python, Streamlit, SQLite, Pandas.

Component Lookup Tool

- Developed a web-based internal tool enabling manufacturing teams to instantly retrieve component details via product IDs, eliminating manual lookup and reducing errors on time-sensitive components (23-hour post-exposure window).
- Tech stack: JavaScript, HTML, CSS.

PROJECTS

PhysioCare — AI-Powered Physiotherapy Rehabilitation Platform — *Flutter, Google ML Kit, MediaPipe BlazePose, Supabase, Gemini 2.5 Flash, Dart PDF, flutter_tts* *2025–2026*

- Led a team of 4 as Team Lead to architect and deliver a cross-platform Flutter rehabilitation app — engineered end-to-end from database schema design to UI layer, driving sprint planning, task delegation, and code reviews throughout the project lifecycle.
- Engineered a dual pose estimation pipeline using Google ML Kit (Android/iOS) and MediaPipe BlazePose (Web) — detecting 33 skeletal landmarks per frame at sub-second latency to evaluate exercise form in real time.
- Developed angle-based rep-counting logic with state machines and cooldown debouncing for 5 exercise types: Bicep Curl, Side Raise, Squats, Standing Hip Abduction, and Seated Knee Extension.
- Integrated live accuracy scoring engine, AI clinical progress notes via Google Gemini 2.5 Flash, downloadable PDF rehabilitation reports, cross-platform Text-to-Speech audio guidance, and a real-time pain alert system notifying therapists and family members.
- Built a full therapist clinical dashboard enabling exercise plan assignment, session report review, patient feedback, and pain alert monitoring — with all session data persisted to Supabase (PostgreSQL).

ComplaintIQ — AI-Powered Banking Complaint Management System — *Python, FastAPI, Streamlit, HuggingFace Transformers, PyTorch, SQLite, Plotly, FPDF, bcrypt* *2025*

- Built a full-stack AI complaint management system for the banking sector that automates ticket classification, prioritization, and resolution using NLP.
- Integrated a PyTorch-based HuggingFace Transformers sentiment analysis pipeline to automatically detect customer sentiment (Angry / Neutral / Happy) and route tickets to the correct department (Loan, Credit Card, Fraud, Account) — reducing manual triage time.
- Implemented urgency detection logic to instantly flag high-priority cases (e.g., fraud), ensuring critical tickets are escalated without human intervention.
- Built a batch processing engine using Pandas to handle CSV uploads and process thousands of complaints simultaneously in seconds.
- Auto-generated professional PDF Executive Summary reports using FPDF for management review, and built an interactive analytics dashboard with Plotly for real-time case metrics.

- Secured the system with bcrypt password hashing, role-based access control (Agent login/registration), and environment variable isolation to prevent credential leaks.

IoT AccessGuard — Smart Attendance & Door Security System — *ESP32, ESP8266, ESP32-CAM, RFID, Arduino IDE, Google Sheets API, Blynk IoT, ML* *2024–2025*

- Engineered a unified IoT product integrating 5 authentication methods — RFID, keypad, fingerprint sensor, face recognition (ML), and IoT mobile app — into a single compact door lock and attendance system.
- Implemented real-time attendance logging via Google Sheets API over Wi-Fi, eliminating manual record-keeping.
- Selected for DIPEX 2025 State-Level Exhibition & Competition, Pune after clearing the Ratnagiri district round — presented to industry and academic judges.

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, SQL, HTML5, CSS3, PHP
AI / ML / NLP: Agentic AI & Multi-Agent Systems (CrewAI, LangGraph, AutoGen/AG2, PydanticAI), LangChain, RAG, ReAct Agents, LLM Integration (Gemini API), PyTorch, NLP & Sentiment Analysis,
Computer Vision (OpenCV, MediaPipe, Google ML Kit, RoboFlow), Custom Dataset Training, Docling
Backend & APIs: FastAPI,Node.js, REST APIs, Python-multipart
Frontend: Streamlit, Flutter, React.js, HTML5, CSS3, Plotly
Data & Databases: Pandas, SQLite, MySQL, Supabase, Google Sheets API, SciPy
IoT & Hardware: ESP32, ESP8266, ESP32-CAM, RFID, Arduino IDE, Blynk IoT
Tools & Platforms: Git, GitHub, VS Code, Google Colab, Postman, npm, Streamlit CLI

EDUCATION

Bachelor of Engineering — Information Technology (Honours in Data Science) *2022 – 2026*
Shree Rayeshwar Institute of Engineering and Information Technology, Shiroda, Goa | **CGPA: 7.61 / 10**

Higher Secondary (HSSC, Class XII) — Goa Board *2022*
Shree Bhumika Higher Secondary School, Goa | **54%**

Secondary (SSC, Class X) — Goa Board *2020*
Vivekanand Dyan Mandir High School, Goa | **74%**

ACHIEVEMENTS & COMPETITIONS

- DIPEX 2025 State-Level Exhibition & Competition, Pune — IoT AccessGuard selected after clearing Ratnagiri district round (organized by ABVP & Srijan Trust).
- Science Film Festival of India (SCIFF 2025) — presented two independent IoT engineering projects to national-level panel.
- CONVERGE 2.0 2025 — participant, organized by Directorate of Higher Education, Goa State Higher Education Council.

CERTIFICATIONS

- **Tata Group Data Analytics Job Simulation** — Forage *(Jan 2026)*
- **IBM SkillsBuild-AI Strategy & Business Intelligence Internship** — AICTE|CSRBOX (April 2026)
- **Agentic AI Hands-on** — IBM Skills Network (1 course, 9 guided projects) *(May 2026)*
- **Applied Artificial Intelligence — AI Agent Development** — CSRBOX Winter Program *(2025)*
- **Data Science Master Virtual Internship** — AICTE-EDUSKILLS *(2024)*
- **AI-ML Virtual Internship** — AICTE-EDUSKILLS *(2024)*
- **Problem Solving through Programming in C** — NPTEL *(2023)*
- **The Joy of Computing Using Python** — NPTEL *(2023)*